ORIGINAL DIGITAL TIMELINE

The Evolution of Artificial Intelligence

From the Turing Test to neural networks, AI winters, cloud computing and generative AI



IDEAS

What can a machine do?

SETBACKS

Why promises outpaced reality

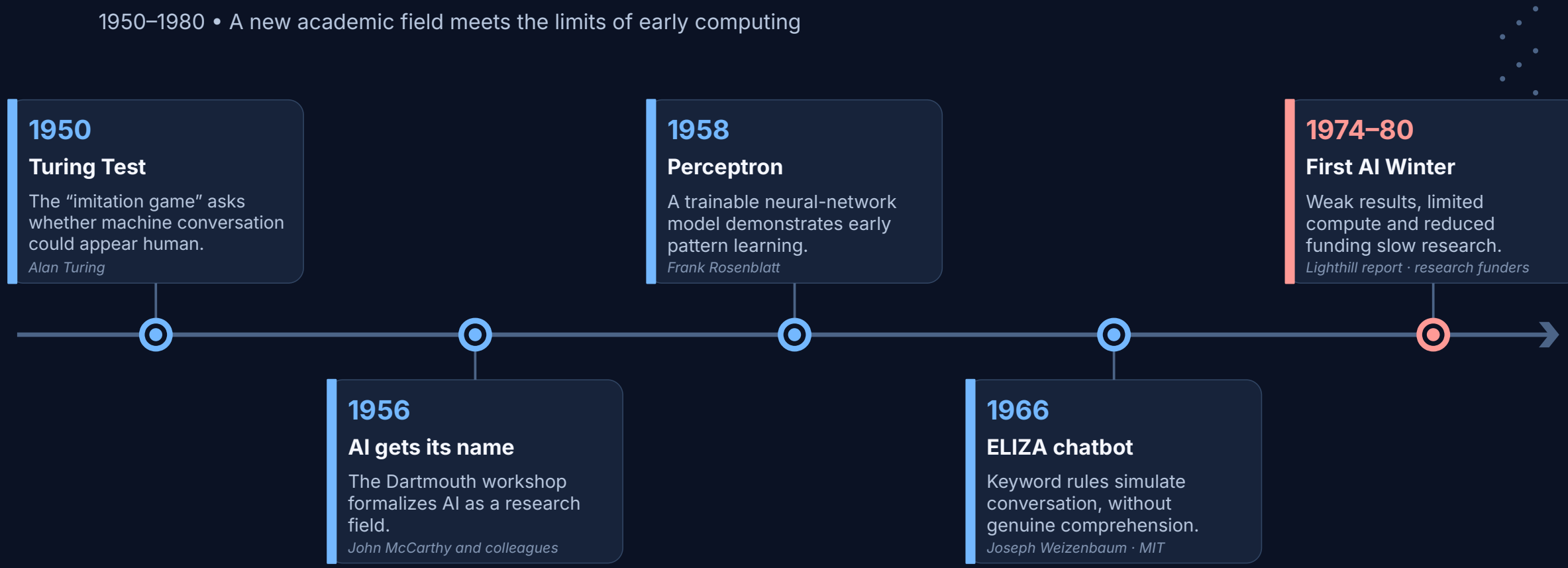
BREAKTHROUGHS

Why data + compute changed AI

01

The Foundations & First Winter

1950–1980 • A new academic field meets the limits of early computing



WHY THIS ERA MATTERS

AI began with promising demonstrations, but early systems struggled outside tightly constrained tasks.

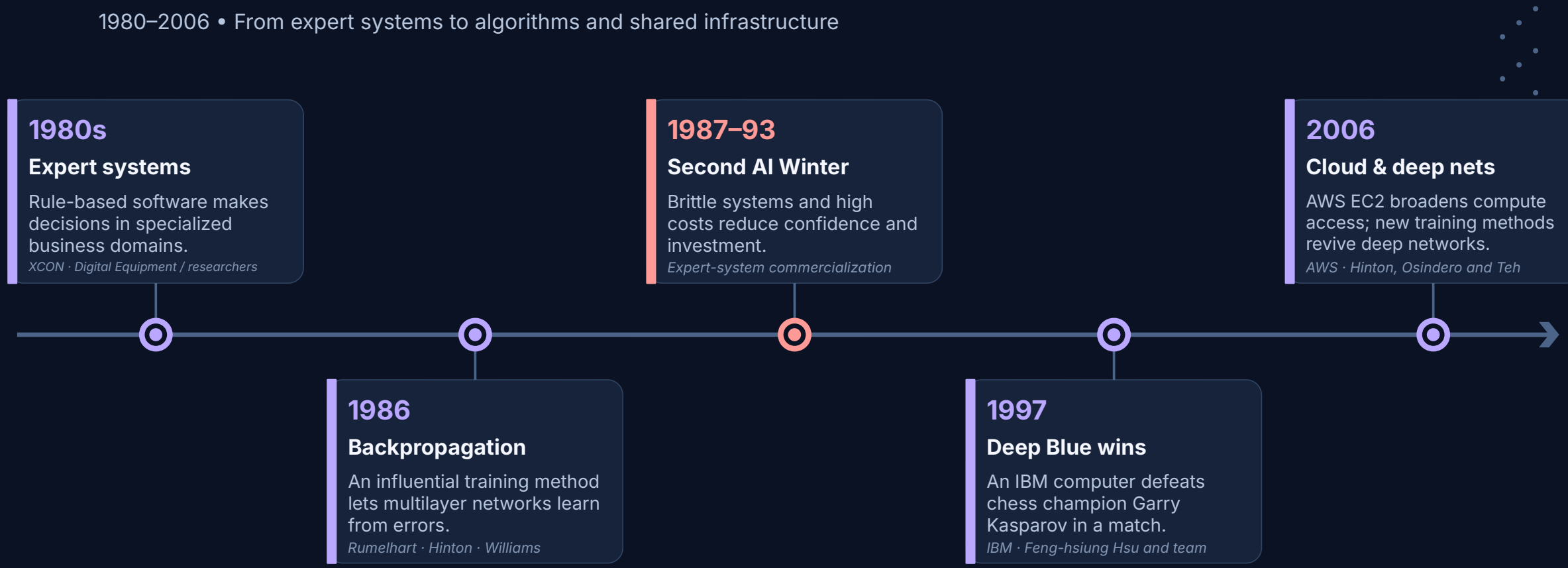
INDUSTRY CONNECTION

Foundations of computing, language processing and automation

02

Rules, Setbacks & a Comeback

1980–2006 • From expert systems to algorithms and shared infrastructure



WHY THIS ERA MATTERS

The 1980s boom exposed the fragility of hand-coded rules; practical learning methods and affordable computing helped AI recover.

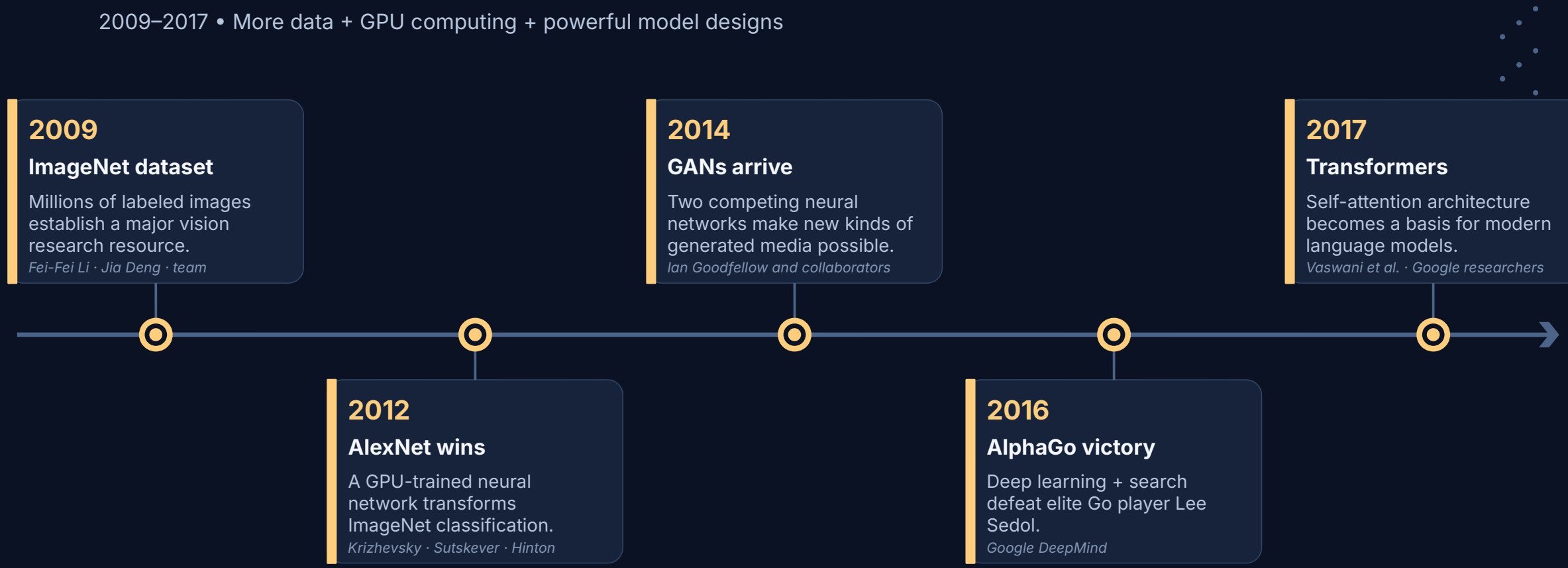
INDUSTRY CONNECTION

Business decision support, finance, logistics and web services

03

The Deep Learning Breakthrough

2009–2017 • More data + GPU computing + powerful model designs



WHY THIS ERA MATTERS

The turning point was not a single invention: datasets, GPUs and better architectures advanced together.

INDUSTRY CONNECTION

Medical imaging, vision, translation, recommendation and speech

04

Generative AI Goes Mainstream

2020–2026 • AI moves from specialist systems into everyday workflows



WHY THIS ERA MATTERS

AI is increasingly useful in daily work, but narrow capabilities, bias, privacy, reliability and energy costs still matter.

INDUSTRY CONNECTION

Health, education, software, science, customer service

05

Research Sources · Part 1

Credible historical and scholarly records • Click blue source labels to open

01 **Bhat, A. K. (2025). The evolution of AI.**
IEEE Computer Society · Open source

02 **Roser, M. (2022). A brief history of artificial intelligence.**
Our World in Data · Open source

03 **Dartmouth. (n.d.). The founding and future of AI.**
Dartmouth College · Open source

04 **Weizenbaum, J. (1966). ELIZA.**
Communications of the ACM · Open source

05 **Guinness World Records. (n.d.). First artificial neural network.**
Guinness World Records · Open source

06 **Actuaries Institute. (2018). History of AI winters.**
Actuaries Institute · Open source

07 **Rumelhart, Hinton & Williams (1986). Back-propagating errors.**
Nature · Open source

08 **Computer History Museum. (n.d.). Deep Blue defeats Kasparov.**
Computer History Museum · Open source

09 **Amazon Web Services. (n.d.). Our origins.**
AWS · Open source

10 **Hinton, Osindero & Teh (2006). Deep belief nets.**
Neural Computation · Open source

06

Research Sources · Part 2

Technical papers, major AI releases and recent adoption research • Click blue labels

11 **Deng et al. (2009). ImageNet: A large-scale image database.**

IEEE CVPR · Open source

12 **Krizhevsky, Sutskever & Hinton (2012). ImageNet classification.**

NeurIPS · Open source

13 **Goodfellow et al. (2014). Generative adversarial networks.**

arXiv · Open source

14 **Google DeepMind. (n.d.). AlphaGo.**

Google DeepMind · Open source

15 **Vaswani et al. (2017). Attention is all you need.**

NeurIPS / Google Research · Open source

16 **OpenAI. (2020). Language models are few-shot learners.**

OpenAI Research · Open source

17 **Google DeepMind. (n.d.). AlphaFold.**

Google DeepMind · Open source

18 **OpenAI. (2022; 2023). Introducing ChatGPT; GPT-4.**

OpenAI · Open source

19 **Stanford HAI. (2026). The 2026 AI Index report.**

Stanford University · Open source

AI ASSISTANCE DISCLOSURE

ChatGPT (GPT-6) assisted with source research, date checking, event summaries and timeline layout. PowerPoint was used to assemble the editable design. Review, personalize and verify before submitting.

Scope: key turning points, not every AI discovery. Dates for AI winters are approximate historical periods.